CHAVVA SAINATH REDDY

SOFTWARE DEVELOPMENT ENGINEER | DATA SCIENTIST & MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

B.Tech CSE (Data Science) from VIT Vellore (CGPA 9.01/10) with strong foundations in Data Structures & Algorithms and object-oriented engineering across C++, Java, and Python. Proven track record building end-to-end deep learning pipelines (EfficientNet, DenseNet, Grad-CAM) and production full-stack web applications with cloud/container architectures.

EDUCATION

Vellore Institute of Technology (VIT), Vellore — **2022 - 2026**
B.Tech in Computer Science and Engineering (Data Science) - CGPA: 9.01 / 10.0

Board of Intermediate Education, Andhra Pradesh (Class XII) — **2020 - 2022**
Higher Secondary - Percentage: 97.1%

Board of Secondary Education, Andhra Pradesh (Class X) — **2020**
Secondary School Certificate - GPA: 10.0 / 10.0

TECHNICAL SKILLS

Languages & Fundamentals: C++, Java, Python, C, JavaScript, SQL, R, DSA, Object-Oriented Design, DBMS, Concurrency
AI, CV & Data Science: TensorFlow, PyTorch, Scikit-learn, OpenCV, DenseNet, EfficientNet, Transfer Learning, Grad-CAM, Pandas, NumPy
Full-Stack & Web: React.js, Node.js, Express.js, Flask, Tailwind CSS, REST APIs, HTML5, CSS3, Bootstrap
Databases & Cloud: MySQL, MongoDB, Oracle DB, Firebase, SQLite, Docker, Git, GitHub Actions, AWS (EC2/S3), GCP, Linux

FEATURED ENGINEERING PROJECTS

Parkinson's Disease Detection from MRI Scans — **Dec 2025 - Jun 2026**
Technologies: TensorFlow, Flask, DeepSeek API, Grad-CAM, Transfer Learning
- Trained CNN architectures (EfficientNetB3, MobileNetV3, ResNet50, DenseNet121) via transfer learning, reaching 99.9% validation accuracy.
- Deployed Flask clinical web application with authentication, instant scan inference, Grad-CAM heatmap overlays, and DeepSeek AI assistant.

Species-Level Malaria Parasite Classifier — **Jul 2025 - Dec 2025**
Technologies: TensorFlow, DenseNet, EfficientNet, OpenCV, Flask
- Differentiated P. falciparum vs P. ovale thin blood smears with 97.0% accuracy overcoming extreme clinical class imbalance.
- Engineered custom image augmentations, weighted loss, Grad-CAM visual attribution, and containerized Flask REST inference service.

Real-Time Bus Route Geospatial Simulation — **Apr 2025**
Technologies: R, Shiny, Leaflet, GPX Spatial Telemetry
- Built interactive telemetry dashboard animating transit vehicles as live moving markers using coordinate GPX logs across VIT-C & Kadapa.
- Dynamically computed real-time speed vectors and responsive route state overlays.

Scalable Food Ordering Platform (Tiffin-Box) — **Apr 2024**
Technologies: Node.js, Express.js, JavaScript, HTML5/CSS3
- Engineered full-stack responsive web system with modular state management, menu browsing, cart checkout, and backend order pipelines.

EXPERIENCE & ACADEMIC RESEARCH

Database Foundations Intern - Oracle Academy — **May 2024 - Jun 2024**
- Executed relational schema design, 3NF normalization, and multi-table SQL queries; earned official Oracle Certification.

Formal Concurrency Verification via Pi-Calculus - Research Project — **Apr 2024**
- Modeled multi-agent distributed systems and mathematically verified deadlock-freeness using Pi-Calculus and SPIN model checker.

CERTIFICATIONS & KEY ACHIEVEMENTS

- IBM Full Stack Software Developer Professional Certificate (15-Course Coursera Specialization: React, Node, DevOps, Docker, AI)
- Oracle Database Foundations Official Certificate (Relational Modeling, SQL, DBMS)
- Competitive Programming: 1,000+ Algorithmic Problems solved across LeetCode, GeeksforGeeks, Coding Ninjas, and HackerRank